Artificial Intelligence & Machine Learning Notes

Foundational study material for an AI/ML RAG PDF Chatbot project

Purpose: This document is designed as a clean, text-based PDF for testing document loading, chunking, embeddings, vector search, and Retrieval-Augmented Generation (RAG).

Contents: AI fundamentals, machine learning, preprocessing, supervised and unsupervised learning, evaluation, deep learning, CNNs, sequence models, NLP, transformers, generative AI, RAG, embeddings, prompt engineering, and responsible AI.

1. Introduction to Artificial Intelligence

Artificial Intelligence (AI) is the field of computer science concerned with building systems that can perform tasks that normally require human intelligence. These tasks include learning, reasoning, perception, language understanding, planning, and decision making.

AI systems can be rule-based, statistical, machine-learning based, or combinations of these approaches. Modern AI relies heavily on data, algorithms, computing power, and mathematical models.

Common applications include recommendation systems, search engines, voice assistants, fraud detection, medical image analysis, autonomous vehicles, robotics, and conversational agents.

2. Intelligent Agents

An intelligent agent perceives its environment through sensors and takes actions through actuators. A rational agent chooses actions that maximize expected performance according to its objective.

Important agent properties include autonomy, reactivity, proactiveness, and the ability to interact with other agents. Examples include a robot navigating a room, a recommendation engine selecting content, and a chatbot responding to users.

Agent environments can be fully or partially observable, deterministic or stochastic, static or dynamic, discrete or continuous, and single-agent or multi-agent.

3. Problem Solving and Search

Many AI problems can be represented as a state space consisting of states, actions, transition rules, a start state, and one or more goal states.

Breadth-First Search explores nodes level by level and can find the shortest solution when every action has the same cost. Depth-First Search explores deeply before backtracking and uses less memory in many cases.

Uniform-Cost Search expands the lowest-cost path. Greedy Best-First Search uses a heuristic estimate, while A* combines path cost and heuristic estimate using $f(n) = g(n) + h(n)$.

4. Knowledge Representation

Knowledge representation describes facts, relationships, rules, and concepts in a form that a computer can process. Common approaches include propositional logic, first-order logic, semantic networks, frames, rules, and ontologies.

A knowledge base stores information and an inference mechanism derives new conclusions. Good representations should support expressiveness, efficient inference, consistency, and easy updating.

5. Machine Learning Fundamentals

Machine Learning (ML) is a subset of AI in which algorithms learn patterns from data rather than being explicitly programmed for every rule.

The major learning paradigms are supervised learning, unsupervised learning, semi-supervised learning, and reinforcement learning. The choice depends on the available data and the problem objective.

A typical ML workflow includes problem definition, data collection, cleaning, exploration, feature engineering, model training, validation, evaluation, deployment, and monitoring.

6. Data Preprocessing

Real-world data often contains missing values, duplicate records, inconsistent formats, outliers, and irrelevant features. Preprocessing improves data quality before modeling.

Common operations include handling missing values, encoding categorical variables, scaling numerical features, removing duplicates, detecting outliers, and selecting useful features.

Feature scaling is particularly important for distance-based and gradient-based algorithms. Standardization transforms a feature using its mean and standard deviation, while normalization can map values to a fixed range.

7. Supervised Learning

Supervised learning uses labeled examples where the input and desired output are known. Regression predicts continuous values, while classification predicts categories.

Examples include predicting house prices, classifying emails as spam or not spam, detecting disease categories, and predicting customer churn.

Common algorithms include linear regression, logistic regression, decision trees, random forests, support vector machines, k-nearest neighbors, and gradient boosting.

8. Linear and Logistic Regression

Linear regression models a continuous target as a weighted combination of input features. Parameters are commonly estimated by minimizing a loss such as mean squared error.

Logistic regression is used for classification. It applies a logistic or sigmoid function to a linear score to produce a probability between zero and one.

Important considerations include feature relationships, multicollinearity, regularization, class balance, and appropriate evaluation metrics.

9. Decision Trees and Ensemble Learning

A decision tree recursively divides data using feature-based rules. It is easy to interpret and can model nonlinear relationships.

Random Forest combines many decision trees trained using randomized samples and features. Aggregating their predictions generally improves robustness and reduces the variance of a single tree.

Boosting builds models sequentially, with later models focusing on errors made by earlier models. Gradient boosting is widely used for structured/tabular data.

10. Unsupervised Learning

Unsupervised learning works with data without target labels. Its goals include discovering groups, patterns, lower-dimensional representations, and unusual observations.

K-means clustering assigns observations to a chosen number of clusters by minimizing within-cluster distance. Hierarchical clustering creates a tree-like structure of nested groups.

Principal Component Analysis (PCA) is a dimensionality-reduction method that transforms correlated features into a smaller set of orthogonal components that capture as much variance as possible.

11. Model Evaluation

For classification, common metrics include accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrices. Accuracy can be misleading when classes are highly imbalanced.

For regression, common metrics include mean absolute error, mean squared error, root mean squared error, and coefficient of determination (R^2).

Cross-validation provides a more reliable estimate of generalization by repeatedly training and validating on different data partitions.

12. Overfitting, Underfitting and Regularization

Overfitting occurs when a model learns training-specific patterns and performs poorly on unseen data. Underfitting occurs when a model is too simple to capture important patterns.

Regularization discourages overly complex models. L1 regularization can encourage sparse parameters, while L2 regularization penalizes large parameter values.

Other techniques include early stopping, dropout in neural networks, data augmentation, feature selection, and collecting more representative data.

13. Neural Networks

A neural network consists of layers of interconnected units. Each unit computes a weighted sum, applies an activation function, and passes the result to the next layer.

Common activation functions include ReLU, sigmoid, and tanh. ReLU is widely used in hidden layers because it is simple and helps reduce some gradient-related problems.

Training usually involves a loss function, forward propagation, backpropagation, and an optimizer such as stochastic gradient descent or Adam.

14. Deep Learning

Deep Learning uses neural networks with multiple layers to learn hierarchical representations. It has achieved strong results in vision, speech, natural language, recommendation, and other areas.

Deep models typically require substantial data and computation. GPUs and specialized accelerators make training practical for many modern applications.

Important concepts include batch size, learning rate, epochs, initialization, normalization, regularization, and transfer learning.

15. Convolutional Neural Networks

Convolutional Neural Networks (CNNs) are especially effective for images. Convolutional filters learn local patterns such as edges, textures, and shapes.

A typical CNN contains convolutional layers, activation functions, pooling or strided operations, and final classification layers. Deeper layers generally represent more complex patterns.

Popular architectures include LeNet, AlexNet, VGG, ResNet, and DenseNet. Residual connections in ResNet help very deep networks train effectively.

16. Recurrent Neural Networks and Sequence Models

Recurrent Neural Networks (RNNs) process sequences while maintaining a hidden state. They have been used for text, speech, and time-series data.

Vanishing and exploding gradients can make long-term dependencies difficult for basic RNNs. LSTM and GRU architectures introduce gating mechanisms to improve sequence learning.

Sequence models can be used for forecasting, classification, language modeling, and sequence-to-sequence tasks.

17. Natural Language Processing

Natural Language Processing (NLP) focuses on enabling computers to work with human language. Tasks include text classification, sentiment analysis, named entity recognition, translation, summarization, question answering, and dialogue.

Traditional NLP pipelines may use tokenization, stop-word handling, stemming or lemmatization, n-grams, TF-IDF, and classical ML models.

Modern NLP commonly uses neural representations and transformer-based architectures that can model relationships between tokens over a context.

18. Transformers and Attention

Attention allows a model to assign different importance to parts of an input when producing an output. Self-attention compares tokens with one another to build contextual representations.

Transformers use attention mechanisms and feed-forward layers rather than recurrence as their primary sequence-processing mechanism. They can process sequences efficiently in parallel during training.

Transformer architectures underpin many modern language models and are also used in vision, audio, and multimodal systems.

19. Generative AI and Large Language Models

Generative AI creates new content such as text, images, audio, code, and synthetic data. Large Language Models (LLMs) are neural models trained on large text corpora to predict and generate language.

An LLM can answer questions, summarize documents, transform text, generate code, and support conversational interfaces. Its output quality depends on training, prompting, context, and model capabilities.

Important limitations include hallucination, bias, outdated knowledge, context limitations, privacy concerns, and sensitivity to prompt formulation.

20. Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) combines information retrieval with language generation. Instead of relying only on the model's internal knowledge, the system retrieves relevant external information and provides it as context.

A typical RAG pipeline is: document ingestion, text extraction, chunking, embedding generation, vector storage, query embedding, similarity search, context construction, and LLM generation.

RAG is useful for private documents, knowledge bases, manuals, policies, research collections, and frequently changing information. Retrieval quality strongly affects final answer quality.

21. Embeddings and Vector Databases

An embedding is a numerical representation of an object such as a sentence, paragraph, image, or document. Semantically similar items tend to have nearby representations in vector space.

Vector databases store embeddings and support similarity search. Common similarity measures include cosine similarity, dot product, and Euclidean distance.

Chroma, FAISS, Pinecone, Weaviate, and Milvus are examples of vector-search technologies. A RAG system typically retrieves the top-k most relevant chunks.

22. Prompt Engineering

Prompt engineering is the practice of designing instructions and context to guide an AI model toward useful and reliable outputs.

Useful prompt components include a clear role, task description, constraints, context, examples when needed, output format, and rules for uncertainty.

For RAG, prompts should instruct the model to ground answers in retrieved context and avoid inventing information that is not supported by the supplied documents.

23. AI Ethics, Safety and Privacy

Responsible AI considers fairness, transparency, accountability, safety, privacy, security, and human oversight.

Bias can enter through data collection, labels, sampling, model design, and deployment decisions. Evaluation should consider different user groups and realistic failure modes.

Sensitive documents should be handled carefully. Systems should minimize unnecessary data collection, protect credentials, apply access controls, and avoid exposing private information in prompts or logs.

24. AI Project Lifecycle

An AI project begins with a clearly defined problem and measurable success criteria. The team then collects suitable data, establishes baselines, trains and evaluates models, and validates the system in realistic conditions.

Deployment requires attention to latency, cost, scalability, reliability, monitoring, versioning, and security. Models may degrade when real-world data changes.

For a RAG chatbot, monitoring can include retrieval relevance, answer quality, latency, token usage, failure rates, and user feedback.